Dynamic rotational feedbacks in the elastic sea-level problem: governing equations and iterative forward solution

Working notes (prepared with Claude) — to be checked

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Abstract

This note is the single reference for the dynamic extension of the rotational sea-level theory of Al-Attar et al. (2024; hereafter AA24). It supersedes `rotation_alattar_notation.pdf` and `rotation_forward_problem.pdf` (hereafter Note 1 and the companion note), whose logic is audited in Appendix A. The earth model is elastic, the ocean is hydrostatic, there is no relative angular momentum apart from the leading-order effect of the fluid core, and no external torque acts. With these assumptions, AA24's static Euler relation between the rotation vector and the inertia perturbation is replaced by the linearised Liouville equation, and nothing else changes. The only memory in the problem is then the equatorial rotation vector, a single 2-vector. The main result is a practical one (§8): the forward problem is solved by marching in time, running the existing static fixed-point iteration at each step with one line changed. An exact superposition method, which needs no change to `pyslfp`, provides an independent check. Iterating over the whole time history at once is shown not to work. Extensions (relative angular momentum, torques, anelasticity) are indicated where they would enter, and the generalised problem and its reciprocity theorem are given in §10. Section 2 shows that for data corrected for the pole tide in the standard way, the load-driven rotation cancels from the observables altogether. The dynamic theory then matters mainly for uncorrected data, for polar motion itself, and for understanding.

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1 Scope, assumptions and notation

The problem. Given a direct load $\zeta(\mathbf{x}, t)$ on $t \in [0, T]$ (for ice, $\zeta = \rho_i(1 - C)\Delta I$), find the sea-level change ξ, displacement $\mathbf{u}$, gravitational potential perturbation ϕ and rotation-vector perturbation $\boldsymbol{\omega}$ as functions of time.

Assumptions.

(A1) **Elastic, quasi-static deformation.** The momentum equation is split by projecting it onto rigid rotations and onto deformations. The rigid-rotation projection, taken in the Tisserand frame, is the Liouville equation, in which all inertial terms are kept (§4). In particular, the Euler force $-\rho\dot{\boldsymbol{\omega}} \times \mathbf{x}$ does no work on Tisserand-frame deformations, since $\int_M \rho\,(\dot{\boldsymbol{\omega}} \times \mathbf{x}) \cdot \mathbf{u}\,\mathrm{d}V = \dot{\boldsymbol{\omega}} \cdot \int_M \rho\,\mathbf{x} \times \mathbf{u}\,\mathrm{d}V = 0$. It therefore enters only through its torque, which is the retained term $\mathbf{I}\dot{\boldsymbol{\omega}}$. The deformational projection is the quasi-static elastic problem: $(\mathbf{u}, \phi)$ respond instantaneously to the current load and centrifugal potential. The inertial and Coriolis forces dropped there are smaller than the elastic restoring force by $(\nu/\omega_0)^2$ and $2(\nu/\Omega)(\Omega/\omega_0)^2 \approx 3 \times 10^{-3}\,\nu/\Omega$, where ω_0 is the frequency of the gravest normal mode ${}_0S_2$.

(A2) **Hydrostatic ocean, fixed shorelines.** The sea surface is at every instant an equipotential of the gravitational plus centrifugal potential, and the ocean function C does not change. The ocean-free problem is the special case $C = 0$.

(A3) **No relative angular momentum**, $\mathbf{h} = \mathbf{0}$, except for the fluid core. Displacements are measured in a Tisserand frame, with no rigid rotation of the solid Earth. Currents, the atmosphere and the transport of meltwater carry no angular momentum relative to that frame. The fluid outer core is different: its relative angular momentum $\mathbf{h}_{\text{core}}$ is driven by the wobble itself. It is small, but its gyroscopic term $\Omega\mathsf{E}\mathbf{h}_{\text{core}}$ is a fraction $A_f/I_1 \approx 0.11$ of the retained $I_1\dot{\boldsymbol{\omega}}$. With the core, the frame is the Tisserand frame of the mantle, and the Euler-force argument of (A1) integrates over the mantle. Its leading-order effect is included through the wobble inertia A_{w} (§4). Setting $A_{\text{w}} = I_1$ recovers the strict assumption $\mathbf{h} = \mathbf{0}$.

(A4) **No external torque**, $\boldsymbol{\Gamma} = \mathbf{0}$.

(A5) **Reference state as in AA24.** The elastic–gravitational operator comes from a spherically symmetric model, and $\mathbf{I}$ is the observed inertia tensor with $\boldsymbol{\Omega} = \Omega\hat{\mathbf{e}}_3$ along the axis of greatest moment. Derivations are for general $I_1 \le I_2 < I_3$. The implementation, like `pyslfp`, uses $I_1 = I_2$.

(A3) and (A4) are relaxed in §9, which shows that any other $\mathbf{h}$, and $\boldsymbol{\Gamma}$, enter only as additional excitation.

Ordering. Let ν be a typical forcing frequency. The theory treats $\nu/\sigma_e = O(1)$ exactly, where $\sigma_e = \Omega(I_3 - I_1)/A_w$ is the Euler frequency, and drops terms of relative size $\epsilon := \nu/\Omega$ and $\delta := (I_3 - I_1)/I_1 \approx 3.3 \times 10^{-3}$. Section 4.2 lists the neglected effects by their actual size. The largest, the fluid core (about 11% of the retained inertial term), is included through A_w. The recommended default configuration is $A_w = A_m$, an equilibrium ocean and damping with $Q \approx 100$.

Notation. The notation follows AA24, with three changes. The background inertia tensor is $\mathbf{I}$ (AA24's $\mathbf{C}$), so the ocean function keeps the symbol C without a clash. Sea-level change is ξ (AA24's ΔSL), as in Heathcote et al. (2026). The gyroscopic matrix is $\mathbf{N}$.

Symbol	Meaning	AA24
$\mathbf{I}$, $I_1 \le I_2 < I_3$	background inertia tensor and principal moments	$\mathbf{C}$, C_i
$\mathbf{c}$	first-order perturbation of the inertia tensor	$\mathbf{c}$
$\mathbf{j} = \mathbf{c}\boldsymbol{\Omega}$	inertia perturbation contracted with $\boldsymbol{\Omega}$	$\mathbf{j}$
$\tilde{\mathbf{I}} = \mathrm{diag}(I_3 - I_1,\, I_3 - I_2,\, -I_3)$	static rotation matrix	$\tilde{\mathbf{C}}$
$\boldsymbol{\Omega} = \Omega\hat{\mathbf{e}}_3$, $\boldsymbol{\omega}$	reference rotation vector and its perturbation	same
Ψ, ψ	centrifugal potential and its perturbation	same
Φ, ϕ	gravitational potential and its perturbation (Eulerian)	same
$\mathbf{u}$	displacement	same
ξ	sea-level change	ΔSL
C	ocean function	C
σ, ζ	total and direct surface loads	same
$\mathbf{k}$	AA24 rotational forcing ($\mathbf{k} = -\mathbf{h}$ statically)	$\mathbf{k}$
$\mathbf{h}$, $\boldsymbol{\Gamma}$	relative angular momentum, external torque	—
$\mathbf{N}$	gyroscopic matrix, eq. (15)	—
A_w; A_f, $A_m = I_1 - A_f$	wobble inertia; fluid-core and mantle equatorial moments	—
$\mathsf{E} = \begin{pmatrix} 0 & -1 \\ 1 & 0 \end{pmatrix}$	$+90°$ rotation of equatorial 2-vectors	—

Subscripts $\perp$ and $\parallel$ denote components orthogonal and parallel to $\hat{\mathbf{e}}_3$. Where no confusion arises, $\boldsymbol{\omega}$ and $\mathbf{j}$ also denote their equatorial 2-vectors (ω_1, ω_2) and (j_1, j_2). Sign conventions are those of AA24. Φ increases outwards, so $\nabla\Phi = g\hat{\mathbf{n}}$ at the surface of a spherical model, and the geoid perturbation is $-\phi/g$.

2 What the observations see

Before building any rotational machinery, it is worth asking which rotational signals survive in the data.

Decomposition. The problem is linear. Any observable can therefore be written as

$$d = d^{\zeta}[\zeta] + \mathsf{X}_d\, \boldsymbol{\omega}_{\perp}, \tag{1}$$

where d^{ζ} is the response to the load with the rotation held at zero (the ocean still responds), and X_d is the full response to a unit prescribed rotation (§8.3). This holds for the static and dynamic theories alike: they differ only in $\boldsymbol{\omega}_{\perp}(t)$.

Pole-tide corrections. Standard processing of GRACE (IERS Conventions; Petit & Luzum 2010), of satellite altimetry (Desai et al. 2015) and of GNSS removes a modelled pole tide. It is computed from the *observed* polar motion $\boldsymbol{\omega}_{\rm obs}$ relative to a conventional mean pole $\bar{\mathbf{m}}(t)$: $d_{\rm corr} = d - \mathsf{X}_d^{\rm corr}(\boldsymbol{\omega}_{\rm obs} - \bar{\mathbf{m}})$. Whatever its cause, $\boldsymbol{\omega}_{\rm obs}$ is the true rotation. If the correction model matches the physics, $\mathsf{X}_d^{\rm corr} = \mathsf{X}_d$, then (noting that IERS y_p is positive towards 90°W, so $\boldsymbol{\omega}_{\rm obs}/\Omega = (x_p, -y_p)$, and that the observed pole differs from $\boldsymbol{\omega}$ by an $O(\epsilon)$ term)

$$\boxed{d_{\rm corr} = d^{\zeta}[\zeta] + \mathsf{X}_d\,\bar{\mathbf{m}}} \tag{2}$$

The load-driven rotation cancels exactly, whether it is static or dynamic.

Consequences.

1. For pole-tide-corrected products, the observation operator for ζ is the *non-rotating* one, plus a known offset from the conventional mean pole. For trend inversions, only $\dot{\bar{\mathbf{m}}}$ enters. Neither the static nor the dynamic rotational theory is then needed for these data.

2. The rotational physics left in corrected data is the model mismatch $(\mathsf{X}_d - \mathsf{X}_d^{\rm corr})(\boldsymbol{\omega}_{\rm obs} - \bar{\mathbf{m}})$. Sources include the anelastic k_2 used by the IERS, the ocean pole-tide model and the mean-pole convention. This mismatch acts mainly on the atmospherically and oceanically excited Chandler and annual wobbles, not on the load-driven part.

3. The dynamic theory is needed for data not corrected for the pole tide (usually tide gauges), for polar motion itself used as data, and for understanding. For uncorrected data, the true rotation also contains excitation that (A3) excludes and the model cannot reproduce.

4. The altimetry convention used in `pyslfp` (Lickley et al. 2018; Heathcote et al. 2026) removes only the direct term ψ/g from modelled sea-surface height. It keeps the load-driven solid-Earth and ocean pole tides, and so matches neither uncorrected nor corrected data.

All of this depends on the conventions of each product: which mean pole is used, and which pole-tide models. These should be settled for the specific data sets before Scheme 1 (§8.2) is used in inversions.

3 The static problem (AA24 §§2–3, restated)

Elastic loading. For a surface load σ, $(\mathbf{u}, \phi)$ satisfy

$$A(\mathbf{u}, \phi \,|\, \mathbf{u}', \phi') + \int_{\partial M} (\mathbf{u}' \cdot \nabla\Phi + \phi')\,\sigma \,\mathrm{d}S = 0 \qquad \text{for all } (\mathbf{u}', \phi'), \tag{3}$$

where the bilinear form A is symmetric under exchange of the pairs $(\mathbf{u}, \phi)$ and $(\mathbf{u}', \phi')$, and vanishes on linearised rigid-body motions (AA24 eqs. 1–4).

Sea level and load. With rotation,

$$\xi = -\frac{1}{g}\,(\mathbf{u} \cdot \nabla\Phi + \phi + \psi) + \frac{\Phi_g}{g}, \qquad \sigma = \zeta + \rho_w C\,\xi, \qquad \int_{\partial M} \sigma \,\mathrm{d}S = 0, \tag{4}$$

where the constant Φ_g is fixed by the mass constraint (AA24 eqs. 10–11, 53).

Centrifugal potential. From $\Psi = -\frac{1}{2}\|\boldsymbol{\Omega} \times \mathbf{x}\|^2$,

$$\psi = -(\boldsymbol{\Omega} \times \mathbf{x}) \cdot (\boldsymbol{\omega} \times \mathbf{x}), \qquad \nabla\psi = \boldsymbol{\Omega} \times (\boldsymbol{\omega} \times \mathbf{x}) + \boldsymbol{\omega} \times (\boldsymbol{\Omega} \times \mathbf{x}). \tag{5}$$

With $\boldsymbol{\Omega} = \Omega\hat{\mathbf{e}}_3$, $\psi = \Omega\,[\,x_3(\omega_1 x_1 + \omega_2 x_2) - \omega_3(x_1^2 + x_2^2)\,]$. The equatorial components of $\boldsymbol{\omega}$ generate a degree-2, order-1 potential. ω_3 generates degree-0 and degree-2, order-0 terms.

Inertia identity. The inertia perturbation produced by a load σ and the resulting displacement $\mathbf{u}$ satisfies, for every constant vector $\boldsymbol{\omega}'$ with associated ψ' (AA24 eqs. 46–49),

$$\boldsymbol{\omega}' \cdot \mathbf{j} + \int_M \rho\, \mathbf{u} \cdot \nabla\psi'\, \mathrm{d}V + \int_{\partial M} \sigma\, \psi'\, \mathrm{d}S = 0. \tag{6}$$

This is a statement about the mass distribution at one instant. It therefore holds unchanged, at each t, in the dynamic problem.

Static rotation. The steady Euler equation $\boldsymbol{\Omega} \times (\mathbf{I}\boldsymbol{\Omega}) = \mathbf{0}$, perturbed to first order, together with conservation of $\|\mathbf{L}\|$, gives $\tilde{\mathbf{I}}\boldsymbol{\omega} = \mathbf{j}$ (AA24 eqs. 36–42). The static problem in weak form is

$$A(\mathbf{u}, \phi \,|\, \mathbf{u}', \phi') + \int_{\partial M} (\mathbf{u}' \cdot \nabla\Phi + \phi')\, \sigma\, \mathrm{d}S + \int_M \rho\, \mathbf{u}' \cdot \nabla\psi\, \mathrm{d}V + (\tilde{\mathbf{I}}\boldsymbol{\omega} - \mathbf{j}) \cdot \boldsymbol{\omega}' = 0, \tag{7}$$

for all $(\mathbf{u}', \phi', \boldsymbol{\omega}')$. Here $-\rho\nabla\psi$ is the centrifugal body force (AA24 eq. 50). AA24's generalised forcings $(\mathbf{t}, \zeta_\phi, \mathbf{k})$ are deferred to §10.1. §4 shows that both halves of $\tilde{\mathbf{I}}\boldsymbol{\omega} = \mathbf{j}$ are limits of the Liouville equation.

4 Rotational dynamics: the linearised Liouville equation

4.1 The full first-order equation

In the Earth-fixed frame, which rotates with $\boldsymbol{\Omega}+\boldsymbol{\omega}$, the angular momentum $\mathbf{L} = (\mathbf{I}+\mathbf{c})(\boldsymbol{\Omega}+\boldsymbol{\omega})+\mathbf{h}$ obeys $\dot{\mathbf{L}} + (\boldsymbol{\Omega}+\boldsymbol{\omega}) \times \mathbf{L} = \boldsymbol{\Gamma}$. Linearising, with $\mathbf{I}\boldsymbol{\Omega} = I_3\boldsymbol{\Omega}$, gives

$$\frac{\mathrm{d}}{\mathrm{d}t}\big(\mathbf{I}\boldsymbol{\omega} + \mathbf{j} + \mathbf{h}\big) + \boldsymbol{\Omega} \times \big(\mathbf{I}\boldsymbol{\omega} + \mathbf{j} + \mathbf{h}\big) + \boldsymbol{\omega} \times (\mathbf{I}\boldsymbol{\Omega}) = \boldsymbol{\Gamma}. \tag{8}$$

In components with general I_1, I_2:

$$\begin{aligned} I_1\dot{\omega}_1 + (I_3 - I_2)\Omega\,\omega_2 &= \Omega(j_2 + h_2) - (\dot{j}_1 + \dot{h}_1) + \Gamma_1, \\ I_2\dot{\omega}_2 - (I_3 - I_1)\Omega\,\omega_1 &= -\Omega(j_1 + h_1) - (\dot{j}_2 + \dot{h}_2) + \Gamma_2, \\ I_3\dot{\omega}_3 &= -(\dot{j}_3 + \dot{h}_3) + \Gamma_3. \end{aligned} \tag{9}$$

With $j_\perp = \Omega(c_{13}, c_{23})$ these are the classical Liouville equations (Munk & MacDonald 1960). The steady, unperturbed case is $\boldsymbol{\Omega} \times (\mathbf{I}\boldsymbol{\Omega}) = \mathbf{0}$.

For $I_1 = I_2$, the equatorial pair takes the compact 2-vector form

$$I_1\dot{\boldsymbol{\omega}} + \dot{\mathbf{j}} + \dot{\mathbf{h}} = \Omega\,\mathsf{E}\big[(I_3 - I_1)\boldsymbol{\omega} - \mathbf{j} - \mathbf{h}\big] + \boldsymbol{\Gamma}, \tag{10}$$

which can be rearranged exactly as

$$\boxed{\dot{\boldsymbol{\omega}} = \sigma_{\mathrm{e}}\,\mathsf{E}\,(\boldsymbol{\omega} - \boldsymbol{\chi}), \qquad \boldsymbol{\chi} = \frac{1}{I_3 - I_1}\Big[\mathbf{j} + \mathbf{h} + \frac{1}{\Omega}\mathsf{E}\big(\boldsymbol{\Gamma} - \dot{\mathbf{j}} - \dot{\mathbf{h}}\big)\Big]} \tag{11}$$

with $\sigma_{\mathrm{e}} = \Omega(I_3 - I_1)/I_1$ for the moment. Here $\boldsymbol{\chi}$ is the (geodetic) excitation. Eqs. (10) and (11) were checked symbolically against eq. (8).

Fluid core and the wobble inertia. A fluid outer core is coupled to the mantle through pressure on the elliptical core–mantle boundary. To leading order in ν/Ω, its relative rotation is $\mathsf{E}\dot{\boldsymbol{\omega}}/\Omega$, which is $O(\epsilon)$ relative to $\boldsymbol{\omega}$: the core almost follows the wobble. Its relative angular momentum in the mantle frame is $\mathbf{h}_{\mathrm{core}} \approx (A_f/\Omega)\mathsf{E}\dot{\boldsymbol{\omega}}$, where A_f is its equatorial moment. The term $-\Omega\mathsf{E}\mathbf{h}_{\mathrm{core}} = A_f\dot{\boldsymbol{\omega}}$ in eq. (10) is then of the same kind as $I_1\dot{\boldsymbol{\omega}}$ and a fraction A_f/I_1 of it. It cannot be ordered away, and $\dot{\mathbf{h}}_{\mathrm{core}}$ is smaller by a further factor ν/Ω. The net effect is to replace

I_1 by the mantle moment $A_m = I_1 - A_f$ in the $\dot{\boldsymbol{\omega}}$ term only (Smith & Dahlen 1981). Write A_w for this *wobble inertia*: $A_\mathrm{w} = I_1$ under the strict form of (A3), and $A_\mathrm{w} = A_m$ with the core. PREM gives $A_f/I_1 = 0.113$, so $A_m/I_1 = 0.887$. From here on,

$$\sigma_\mathrm{e} := \frac{\Omega(I_3 - I_1)}{A_\mathrm{w}}, \tag{12}$$

with period 304.5 d for $A_\mathrm{w} = I_1$ and 270.1 d for $A_\mathrm{w} = A_m$. The static limit does not involve A_w and is unchanged. The Love numbers relevant to the wobble also differ slightly when the core is decoupled (Smith & Dahlen 1981). That effect is neglected here.

4.2 Ordering, and the leading-order equation

Scale time by the Euler frequency, $\tau = \sigma_\mathrm{e} t$. With $\mathbf{h} = \boldsymbol{\Gamma} = \mathbf{0}$ (the core now being carried by A_w), eq. (11) becomes

$$\frac{\mathrm{d}\boldsymbol{\omega}}{\mathrm{d}\tau} = \mathsf{E}(\boldsymbol{\omega} - \boldsymbol{\chi}), \qquad \boldsymbol{\chi} = \frac{1}{I_3 - I_1}\Big[\mathbf{j} - \delta_w\, \mathsf{E}\frac{\mathrm{d}\mathbf{j}}{\mathrm{d}\tau}\Big], \qquad \delta_w := \frac{\sigma_\mathrm{e}}{\Omega} = \frac{I_3 - I_1}{A_\mathrm{w}}, \tag{13}$$

with $\delta_w = 3.3 \times 10^{-3}$ for $A_\mathrm{w} = I_1$ and 3.7×10^{-3} for $A_\mathrm{w} = A_m$. In the natural limit $\nu/\sigma_\mathrm{e} = O(1)$, the rotation responds on the forcing timescale. For $A_\mathrm{w} = I_1$, ν/σ_e is 10.2 at a period of one month, 0.83 at one year and 0.08 at ten years. The only small parameter left is δ_w. The $\mathrm{d}\mathbf{j}/\,\mathrm{d}\tau$ term is a uniform correction to the excitation of relative size $\delta_w\, \nu/\sigma_\mathrm{e} = \nu/\Omega = \epsilon$: about 3% at monthly periods and 0.3% at annual periods. It is dropped, as in the standard geodetic excitation functions (Barnes et al. 1983; Gross 2015). Keeping it would be harmless but not systematically better, because the relative angular momentum of the transported meltwater, excluded by (A3), is of the same order.

The **leading-order equations** are

$$\boxed{\dot{\boldsymbol{\omega}}_\perp = \sigma_\mathrm{e}\, \mathsf{E}\,(\boldsymbol{\omega}_\perp - \mathbf{q}), \qquad \mathbf{q} := \frac{\mathbf{j}_\perp}{I_3 - I_1}; \qquad\qquad I_3\, \omega_3 = -j_3} \tag{14}$$

with σ_e from eq. (12). The axial relation follows from the third of eqs. (9) integrated from a quiescent start. It is algebraic and carries no memory, which is why AA24's use of $\|\mathbf{L}\|$ conservation is exact at this order. (On sub-decadal timescales the core is also largely decoupled axially, which would replace I_3 by the mantle's polar moment. This is not pursued here, since ω_3 is neglected in the implementation.)

Neglected effects, by size.

- *Anelasticity.* It lengthens the Chandler period by about 8 d (2%) and gives $Q \approx 100$ (Smith & Dahlen 1981). It can be represented by the damping of §8.1.
- *Non-equilibrium ocean pole tide.* Its size is set by ocean dynamics, not by ν/Ω. The observed pole tide is close to the self-consistent equilibrium response (Desai 2002).
- $\dot{\mathbf{j}}$ *and the meltwater* $\mathbf{h}$*:* $O(\epsilon)$, as above.
- *Ellipticity of the elastic response*, i.e. spherically symmetric Love numbers: $O(\delta)$, since the flattening $1/298 \approx \delta$. This is comparable to the $O(\epsilon)$ terms near the annual band ($\epsilon = \delta$ at periods of about 300 d).
- *Solid-Earth inertial and Coriolis forces in the deformation*: $(\nu/\omega_0)^2$ and $3 \times 10^{-3}\epsilon$ (A1). These are negligible.

If $A_\mathrm{w} = I_1$ is used, the fluid core ($\approx 11\%$ of the retained term) heads this list.

What controls the response. The dimensionless forcing frequency that matters for the response is $\nu/\sigma_{\rm CW}$, not $\nu/\sigma_{\rm e}$, where $\sigma_{\rm CW}$ is the free (Chandler) frequency of §6. At one year $\nu/\sigma_{\rm CW}$ is 1.30 ($A_{\rm w} = I_1$) or 1.16 ($A_{\rm w} = A_m$). The annual band therefore lies just beyond the resonance, which is why the annual factors in §7 are large and negative.

Static limit. Setting $\dot{\boldsymbol{\omega}} = \mathbf{0}$ gives $\boldsymbol{\omega}_\perp = \mathbf{q}$, i.e. $(I_3 - I_1)\boldsymbol{\omega}_\perp = \mathbf{j}_\perp$, which is AA24's eq. 38. The static and dynamic theories share the same $\mathbf{q}$. The only difference is that the static theory *assigns* $\boldsymbol{\omega}_\perp = \mathbf{q}$, whereas the dynamic theory makes $\boldsymbol{\omega}_\perp$ *relax towards* $\mathbf{q}$ by circling it (§7).

Matrix form. For general I_1, I_2, eqs. (9) at leading order are

$$\tilde{\mathbf{I}}\boldsymbol{\omega} + \mathbf{N}\dot{\boldsymbol{\omega}} = \mathbf{j}, \qquad \mathbf{N} = \frac{1}{\Omega}\begin{pmatrix} 0 & -A_{w,2} & 0 \\ A_{w,1} & 0 & 0 \\ 0 & 0 & 0 \end{pmatrix}, \tag{15}$$

where $A_{w,i} = I_i$, or the mantle moments with the core. Without the core, $\mathbf{N}\mathbf{v} = \Omega^{-2}\boldsymbol{\Omega} \times (\mathbf{I}\mathbf{v})$. $\mathbf{N}$ has no axial part, so the axial row is exactly $-I_3\omega_3 = j_3$. $\mathbf{N}$ is antisymmetric if and only if $A_{w,1} = A_{w,2}$, in which case its equatorial block is $(A_{\rm w}/\Omega)\mathsf{E}$. The antisymmetry matters only for reciprocity (§10). The forward solver works equally well for $I_1 \neq I_2$.

5 The dynamic forward problem

Collecting eqs. (4), (6), (7) and (15): for each $t \in [0, T]$ and all test functions $(\mathbf{u}', \phi', \boldsymbol{\omega}')$,

$$\boxed{A(\mathbf{u}, \phi \,|\, \mathbf{u}', \phi') + \int_{\partial M} (\mathbf{u}' \cdot \nabla\Phi + \phi')\,\sigma \,\mathrm{d}S + \int_M \rho\, \mathbf{u}' \cdot \nabla\psi \,\mathrm{d}V + (\tilde{\mathbf{I}}\boldsymbol{\omega} + \mathbf{N}\dot{\boldsymbol{\omega}} - \mathbf{j}) \cdot \boldsymbol{\omega}' = 0} \tag{16}$$

with $\boldsymbol{\omega}(0) = \boldsymbol{\omega}_0$, where $\sigma = \zeta + \rho_w C\xi$, ξ is given by eq. (4), $\int_{\partial M} \sigma \,\mathrm{d}S = 0$, and $\mathbf{j}$ is the inertia perturbation of the current configuration.

Structure. Everything in eq. (16) except $\mathbf{N}\dot{\boldsymbol{\omega}}$ is instantaneous. If $\boldsymbol{\omega}(t)$ were known, the fields at each t would follow from a static fingerprint problem with the rotation vector *prescribed*, i.e. with no rotation update. The problem therefore splits into

(i) an instantaneous map $(\zeta(t), \boldsymbol{\omega}(t)) \mapsto (\xi, \mathbf{u}, \phi, \mathbf{q})(t)$, which is the static solver with $\boldsymbol{\omega}$ held fixed; and

(ii) the 2-vector ODE $\dot{\boldsymbol{\omega}}_\perp = \sigma_{\rm e}\mathsf{E}(\boldsymbol{\omega}_\perp - \mathbf{q})$, plus the algebraic ω_3.

The whole memory of the system is $\boldsymbol{\omega}_\perp(t)$. Every method in §8 is a way of coupling (i) and (ii).

Initial state. Take $t = 0$ as the reference epoch, with $\zeta(0) = 0$ and $\boldsymbol{\omega}_0 = \mathbf{0}$. If a load ζ_0 has been in place long enough for equilibrium, instead set $\boldsymbol{\omega}_0$ to the static solution for ζ_0, which also gives $\mathbf{q}_0 = \boldsymbol{\omega}_0$. Starting with $\zeta(0) \neq 0$ and $\boldsymbol{\omega}_0 = \mathbf{0}$ is inconsistent: it excites a spurious free wobble whose amplitude equals the static response to ζ_0. A free wobble genuinely present at $t = 0$ is not determined by the loads, and $\boldsymbol{\omega}_0$ must then be prescribed.

6 Spherically symmetric implementation

6.1 Love numbers and the rotational terms

With $\sigma = \sum \sigma_{lm} Y_{lm}$ (orthonormal real harmonics, no Condon–Shortley phase), AA24 eq. 60 gives $u_{lm} = h_l \sigma_{lm}$ and $\phi_{lm} = k_l \sigma_{lm}$. Here AA24's k_l includes the potential of the load itself. The

centrifugal perturbation ψ acts on the Earth like a tidal potential. The deformation it causes adds $h_2^T\psi_{21}$ to the $(2,1)$ coefficient of the displacement and $k_2^T\psi_{21}$ to that of ϕ, where h_2^T and k_2^T are tidal Love numbers. Here ϕ is the potential of the Earth's redistributed mass only and does not contain ψ itself. This differs from AA24's loading k_l, which includes the potential of the load. Writing $\boldsymbol{\phi}_{21} = (\phi^c_{21}, \phi^s_{21})$ at radius b (the mean sea-floor radius in `pyslfp`):

$$\boldsymbol{\psi}_{21} = r\,\boldsymbol{\omega}_\perp, \quad r = \sqrt{\tfrac{4\pi}{15}}\,\Omega b^2; \qquad \mathbf{q} = \beta\,\boldsymbol{\phi}_{21}, \quad \beta = \sqrt{\tfrac{5}{12\pi}}\,\frac{\Omega b^3}{G(I_3 - I_1)}. \tag{17}$$

The first relation follows from $\psi = \Omega x_3(\omega_1 x_1 + \omega_2 x_2)$ and $Y^c_{21} = \sqrt{15/4\pi}\,x_1x_3/r^2$. The second is MacCullagh's formula: the exterior degree-2 potential of the perturbed mass has $\phi^c_{21} = \sqrt{12\pi/5}\,Gc_{13}/b^3$, and similarly for ϕ^s_{21} with c_{23}, while $\mathbf{j}_\perp = \Omega(c_{13}, c_{23})$. Here $\boldsymbol{\phi}_{21}$ is the potential of *all* perturbed mass (load, internal deformation, and the tidal response to ψ) but not ψ itself. It therefore contains both terms of eq. (6) automatically. In `pyslfp`, r and β are `rotation_factor` and `inertia_factor`, and the static update is `angular_velocity_change = i * potential_change_lm.coeffs[:, 2, 1]`, i.e. $\boldsymbol{\omega}_\perp \leftarrow \mathbf{q}$.

6.2 The rotational feedback matrix

Because map (i) is linear,

$$\mathbf{q}(t) = \mathbf{q}^\zeta(t) + \mathsf{K}\,\boldsymbol{\omega}_\perp(t). \tag{18}$$

Here $\mathbf{q}^\zeta$ is the excitation from the direct load, with the ocean responding and $\boldsymbol{\omega} \equiv \mathbf{0}$. K is a constant 2×2 matrix giving the excitation raised by a unit rotation, through the solid Earth's tidal yielding and the equilibrium ocean pole tide. With no ocean, $\mathsf{K} = \kappa\mathsf{1}$ with

$$\kappa = \beta r k_2^T = \frac{k_2^T}{k_s}, \qquad k_s = \frac{3G(I_3 - I_1)}{\Omega^2 b^5}. \tag{19}$$

General form of K. A unit rotation raises $\boldsymbol{\psi}_{21} = r\hat{\mathbf{e}}_\alpha$. That changes the degree-2, order-1 inertia by two routes. The solid Earth yields tidally, adding $k_2^T\boldsymbol{\psi}_{21}$ to $\boldsymbol{\phi}_{21}$; this is the $\kappa\mathsf{1}$ above. The ocean follows the rotational equipotential, and the degree-2, order-1 part of the resulting water load adds $k_2\boldsymbol{\sigma}_{21}$. The forcing sea level is $-\gamma_2\psi/g$ plus a constant, where $\gamma_2 = 1 + k_2^T - h_2^T \approx 0.69$ is the usual diminishing factor (dimensionless Love numbers). Hence

$$\mathsf{K} = \kappa\mathsf{1} + \mathsf{K}_{\mathrm{oc}}, \qquad \mathsf{K}_{\mathrm{oc}} = \frac{1}{k_s}\,\frac{3\rho_w}{5\bar{\rho}}\,(1 + k_2')\,\gamma_2\,\mathsf{Q}, \qquad \mathsf{Q}_{\alpha\beta} = \int C\,Y^\alpha_{21}\,\hat{\xi}^\beta\,\mathrm{d}\Omega, \tag{20}$$

where k_2' is the degree-2 load Love number, $\bar{\rho}$ the mean density, and the integral runs over the unit sphere. Here $\hat{\xi}^\beta$ is the sea level produced by a unit forcing Y^β_{21}, with the ocean's self-attraction and loading and with mass conserved. Without self-attraction and loading, $\mathsf{Q}_{\alpha\beta} = \int CY^\alpha Y^\beta\,\mathrm{d}\Omega - (\int CY^\alpha\,\mathrm{d}\Omega)(\int CY^\beta\,\mathrm{d}\Omega)/A_{\mathrm{oc}}$, which measures how much the two harmonics overlap over the ocean. The second term is the mass-conservation correction. K_{oc} is positive definite and anisotropic because the ocean geometry is. For present-day oceans this overlap matrix is $\left(\begin{smallmatrix} 0.785 & -0.021 \\ -0.021 & 0.621 \end{smallmatrix}\right)$, and the prefactor is 0.055. Together they reproduce the off-diagonal of K_{oc} in eq. (21) below almost exactly. On the diagonal, self-attraction and loading amplify it by 19%.

K is symmetric. Let $(\mathbf{u}^a, \phi^a, \sigma^a)$ and $(\mathbf{u}^b, \phi^b, \sigma^b)$ be the responses to prescribed rotations $\boldsymbol{\omega}^a, \boldsymbol{\omega}^b$ with no direct load. Using the static weak form without its rotation row, once with each solution as the test function, and using the symmetry of A:

$$\int_{\partial M}(\mathbf{u}^b\cdot\nabla\Phi + \phi^b)\sigma^a\,\mathrm{d}S + \int_M \rho\,\mathbf{u}^b\cdot\nabla\psi^a\,\mathrm{d}V = \int_{\partial M}(\mathbf{u}^a\cdot\nabla\Phi + \phi^a)\sigma^b\,\mathrm{d}S + \int_M \rho\,\mathbf{u}^a\cdot\nabla\psi^b\,\mathrm{d}V.$$

Eq. (6) then gives

$$\boldsymbol{\omega}^b \cdot \mathbf{j}^a - \boldsymbol{\omega}^a \cdot \mathbf{j}^b = \int_{\partial M} (\mathbf{u}^a \cdot \nabla\Phi + \phi^a + \psi^a)\sigma^b \, \mathrm{d}S - \int_{\partial M} (\mathbf{u}^b \cdot \nabla\Phi + \phi^b + \psi^b)\sigma^a \, \mathrm{d}S.$$

Substitute eq. (4) with $\sigma = \rho_w C\xi$. The constants Φ_g drop out by mass conservation, and what remains is $-g\rho_w \int C\xi^a\xi^b + g\rho_w \int C\xi^b\xi^a = 0$. Hence $\boldsymbol{\omega}^b \cdot \mathbf{j}^a = \boldsymbol{\omega}^a \cdot \mathbf{j}^b$, and with $\mathbf{j}^a = (I_3 - I_1)\mathsf{K}\boldsymbol{\omega}^a$, $\mathsf{K} = \mathsf{K}^T$. For $I_1 \neq I_2$ the same argument shows that the matrix taking $\boldsymbol{\omega}_\perp$ to $\mathbf{j}_\perp$ is symmetric.

Numbers. The `pyslfp` constants give $k_s = 0.951$ and, with $k_2^T = 0.2986$, $\kappa = 0.314$. `pyslfp` takes $\Omega = 2\pi/86400\,\mathrm{s}^{-1}$ (solar day). The sidereal $7.2921 \times 10^{-5}\,\mathrm{s}^{-1}$ should be used instead. It lowers k_s by 0.55%, raises κ and K by the same factor, and shortens the Euler period by 0.27%, but it changes the Chandler periods below by less than 0.3 d. The numbers here use the `pyslfp` constants. Two static `pyslfp` solves at $l_{\max} = 64$ with present-day oceans (the method is in §8.3) give

$$\mathsf{K} = \begin{pmatrix} 0.3655 & -0.0012 \\ -0.0012 & 0.3550 \end{pmatrix}, \tag{21}$$

which is symmetric to 2×10^{-15}. The ocean pole tide therefore adds about 0.04–0.05 to the solid-Earth κ, anisotropically.

6.3 Free wobble

Substituting eq. (18) into eq. (14),

$$\dot{\boldsymbol{\omega}}_\perp = \sigma_\mathrm{e}\,\mathsf{E}\big[(1 - \mathsf{K})\boldsymbol{\omega}_\perp - \mathbf{q}^\zeta\big]. \tag{22}$$

Because K is symmetric, $\mathsf{E}(1-\mathsf{K})$ has zero trace. Its eigenvalues therefore satisfy $\lambda^2 = -\det(1-\mathsf{K})$, and because $1 - \mathsf{K}$ is positive definite they are purely imaginary, $\pm i\sigma_\mathrm{CW}$, with

$$\sigma_\mathrm{CW} = \sigma_\mathrm{e}\sqrt{\det(1 - \mathsf{K})}. \tag{23}$$

T_CW	no ocean	equilibrium ocean
$A_\mathrm{w} = I_1$ (no core)	444.0 d	476.0 d
$A_\mathrm{w} = A_m$ (fluid core)	393.7 d	422.2 d

The equilibrium ocean lengthens the period by about 30 d (Dahlen 1976), and the core shortens it by 11%. The free motion is undamped and, because K is slightly anisotropic, slightly elliptical. Smith & Dahlen (1981) obtained 426.7 sidereal days (425.5 d) for an elastic Earth with a fluid core and an equilibrium ocean. The observed period is about 433 d. So $A_\mathrm{w} = A_m$ is within 1% of the complete elastic value, and the remaining 8 d is attributed to anelasticity.

7 What the equation does: analytic solutions for testing

The solutions below come from eq. (22) with $\mathsf{K} = \kappa\mathsf{1}$ and no damping, or from its complex form $m = \omega_1 + i\omega_2$, in which E becomes multiplication by i. They double as test cases for a solver.

Constant excitation. With $\boldsymbol{\omega}_s := (1 - \mathsf{K})^{-1}\mathbf{q}^\zeta$ the static answer, $\boldsymbol{\omega}_\perp(t) = \boldsymbol{\omega}_s + \exp[\sigma_\mathrm{e}\mathsf{E}(1 - \mathsf{K})t]\,(\boldsymbol{\omega}_0 - \boldsymbol{\omega}_s)$. The rotation vector circles the static answer at the Chandler rate, prograde, and never settles. A load switched on at $t = 0$ from rest excites a free wobble whose radius equals the static response.

Periodic excitation. For $\mathbf{q}^\zeta \propto e^{i\nu t}$ ($\nu > 0$ prograde, $\nu < 0$ retrograde),

$$\frac{m_{\text{dynamic}}}{m_{\text{static}}} = \frac{1}{1 - \nu/\sigma_{\text{CW}}}, \tag{24}$$

which is a resonance on the prograde side only (isotropic approximation, using each configuration's T_{CW}):

Forcing period	30 d	182.6 d	365.25 d	$\to \infty$
prograde, $T_{\text{CW}} = 476$ d ($A_{\text{w}} = I_1$)	-0.067	-0.62	-3.30	1
retrograde, $T_{\text{CW}} = 476$ d	0.059	0.28	0.43	1
prograde, $T_{\text{CW}} = 422$ d ($A_{\text{w}} = A_m$)	-0.077	-0.76	-6.42	1
retrograde, $T_{\text{CW}} = 422$ d	0.066	0.30	0.46	1

At monthly periods the polar motion is an order of magnitude smaller than static theory predicts and of opposite sign. At annual periods the prograde part is amplified several-fold, by an amount sensitive to T_{CW}: -5.4 at the observed 433 d. These factors multiply the rotational *correction* to sea level, not sea level itself.

Linear trend. For $\mathbf{q}^\zeta = \dot{\mathbf{q}}^\zeta t$ from rest,

$$\boldsymbol{\omega}_\perp(t) = \boldsymbol{\omega}_s(t) + \frac{1}{\sigma_{\text{CW}}}\,\mathsf{E}^T\,\dot{\boldsymbol{\omega}}_s + \boldsymbol{\omega}_{\text{free}}(t), \qquad \boldsymbol{\omega}_s(t) = \frac{\dot{\mathbf{q}}^\zeta t}{1-\kappa}, \tag{25}$$

where $\boldsymbol{\omega}_{\text{free}}$ is a free wobble of the same amplitude as the constant offset. The trend is exactly the static one. The offset $\mathsf{E}^T\dot{\boldsymbol{\omega}}_s/\sigma_{\text{CW}}$ is *perpendicular* to the trend: the pole is displaced sideways from the static path by the distance the static pole travels in $T_{\text{CW}}/2\pi$. That is 76 d for $T_{\text{CW}} = 476$ d and 67 d for 422 d. It is not a lag along the path, which would be parallel to the trend. Decadal trends are therefore unaffected, while sub-annual variability is.

8 Solving the forward problem iteratively

8.1 Time discretisation

The elastic theory has no dissipation, so its free wobble never decays. The real Chandler wobble has $Q \approx 100$, mainly from mantle anelasticity. The solver therefore integrates

$$\dot{\boldsymbol{\omega}}_\perp = \mathsf{L}\,(\boldsymbol{\omega}_\perp - \mathbf{q}), \qquad \mathsf{L} = \sigma_{\text{e}}\mathsf{E} - \gamma\mathsf{1}, \tag{26}$$

where $\gamma \geq 0$ is a phenomenological damping of the Munk–MacDonald type. It leaves the static limit unchanged. With $\mathsf{K} = \kappa\mathsf{1}$ the free mode is $e^{(i\sigma_{\text{CW}} - \gamma(1-\kappa))t}$, so $Q = \sigma_{\text{e}}/2\gamma$. The recommended default for applications is $Q \approx 100$. Setting $\gamma = 0$ recovers the purely elastic theory, which is used for the checks of §8.5. Damping preserves reciprocity (§10.3).

Apply the trapezoidal rule on a grid $t_n = n\Delta t$, with $\boldsymbol{\omega}_n = \boldsymbol{\omega}_\perp(t_n)$, $\mathbf{q}_n = \mathbf{q}(t_n)$ and $h = \Delta t/2$:

$$\boxed{\boldsymbol{\omega}_{n+1} = \mathsf{R}\,\boldsymbol{\omega}_n + \mathsf{S}\,(\mathbf{q}_n + \mathbf{q}_{n+1}), \qquad \mathsf{R} = (1 - h\mathsf{L})^{-1}(1 + h\mathsf{L}), \qquad \mathsf{S} = -(1 - h\mathsf{L})^{-1}h\mathsf{L}} \tag{27}$$

For $\gamma = 0$ and $a = \sigma_{\text{e}}\Delta t/2$, these are $\mathsf{R} = [(1-a^2)\mathsf{1} + 2a\mathsf{E}]/(1+a^2)$ and $\mathsf{S} = (a^2\mathsf{1} - a\mathsf{E})/(1+a^2)$. Properties:

- $\mathsf{R} + 2\mathsf{S} = \mathsf{1}$, so a steady state $\boldsymbol{\omega} = \mathbf{q}$ is preserved exactly. For $\gamma = 0$, R is an exact rotation by $2\arctan a$, so the scheme adds no numerical damping of its own.
- $\|\mathsf{S}\| < 1$ at every Δt, with $\|\mathsf{S}\| = a/\sqrt{1+a^2}$ for $\gamma = 0$. This follows because L is normal with eigenvalues of non-positive real part. It is what makes the inner iteration of §8.2 converge unconditionally.

- **Accuracy.** With feedback, the discrete free mode is a rotation by $2\arctan(a_{\mathrm{CW}})$ per step, $a_{\mathrm{CW}} = \sigma_{\mathrm{CW}}\Delta t/2$, so the period is too long by about $a_{\mathrm{CW}}^2/3$: 0.09% for weekly steps and 1.7% for monthly steps (429.3 against 422.2 d in the default configuration). The phase error accumulates over a run. In a 20-year test with weekly steps the maximum error in $\boldsymbol{\omega}$ was 3.5%. For long runs, use Δt of a week or less, or use the exact propagator of §8.3.
- Backward Euler, $\boldsymbol{\omega}_{n+1} = [(1 + b\mathsf{E})\boldsymbol{\omega}_n + (b^2\mathsf{1} - b\mathsf{E})\mathbf{q}_{n+1}]/(1 + b^2)$ with $b = \sigma_{\mathrm{e}}\Delta t$ (for $\gamma = 0$), damps the wobble numerically but tends to the static solution as $\Delta t \to \infty$. It is useful only as a check against the static solver.
- **Units.** `pyslfp` is non-dimensional. Compute σ_{e} from `rotation_frequency`, the two moments and $A_f/I_1 = 0.113$ (PREM; not in `pyslfp`), and express Δt in units of `time_scale`.

8.2 Scheme 1: march in time with an inner fixed-point iteration (recommended)

The static solver already iterates between the load, the fields and $\boldsymbol{\omega}$. Only its rotation update changes. Given the converged $\boldsymbol{\omega}_n$ and $\mathbf{q}_n$ from the previous step and the load ζ_{n+1}:

1. Initialise $\boldsymbol{\omega}^{(0)} = \boldsymbol{\omega}_n$ and the load $\sigma^{(0)}$ (a warm start from the converged σ_n is better than a cold one).
2. For $k = 0, 1, \ldots$ until the load converges:
   (a) $\boldsymbol{\psi}_{21} = r\,\boldsymbol{\omega}^{(k)}$;
   (b) $u_{lm} = h_l\sigma_{lm}^{(k)}$, $\phi_{lm} = k_l\sigma_{lm}^{(k)}$, adding $h_2^T\boldsymbol{\psi}_{21}$ and $k_2^T\boldsymbol{\psi}_{21}$ at $(2, 1)$;
   (c) $\mathbf{q}^{(k)} = \beta\boldsymbol{\phi}_{21}$;
   (d) **the only change:** $\boldsymbol{\omega}^{(k+1)} = \mathsf{R}\boldsymbol{\omega}_n + \mathsf{S}(\mathbf{q}_n + \mathbf{q}^{(k)})$, in place of the static $\boldsymbol{\omega}^{(k+1)} = \mathbf{q}^{(k)}$;
   (e) add $r\boldsymbol{\omega}^{(k+1)}$ to $\boldsymbol{\phi}_{21}$, form ξ from eq. (4), and set $\sigma^{(k+1)} = \zeta_{n+1} + \rho_w C\xi$.
3. Store $\boldsymbol{\omega}_{n+1} = \boldsymbol{\omega}^{(k+1)}$ and $\mathbf{q}_{n+1} = \mathbf{q}^{(k)}$, the latter computed with ϕ excluding ψ.

In `pyslfp` this is a copy of `solve_generalised_equation` with the rotation update replaced:

```
q = i * potential_change_lm.coeffs[:, 2, 1]
if angular_momentum_change is not None:
    q -= m * angular_momentum_change
angular_velocity_change = R @ w_old + S @ (q_old + q)
```

The copy takes `w_old` and `q_old` as inputs and returns the converged `q`. The `angular_momentum_change` lines carry the generalised forcing $\mathbf{k}$ of §10.1 and are inactive in a forward run. The static solver is the case $\mathsf{R} = 0$, $\mathsf{S} = \mathsf{1}$ with the $\mathbf{q}_n$ term removed.

Convergence. With the ocean converged, the rotational part of the inner iteration is $\boldsymbol{\omega} \mapsto \mathsf{R}\boldsymbol{\omega}_n + \mathsf{S}(\mathbf{q}_n + \mathbf{q}_{n+1}^{\zeta} + \mathsf{K}\boldsymbol{\omega})$. It contracts by $\|\mathsf{SK}\| \le \|\mathsf{S}\|\,\kappa_{\max} < 0.37$ at every step size and forcing frequency, and more strongly for small steps. In a test with the K of eq. (21) and weekly steps, the iteration reached 10^{-14} in about 10 iterations. In practice the ocean-load iteration, which is the same as in the static solver, dominates the iteration count.

Why this is the recommended solver. It is the smallest change to working code. It never iterates over time: the problem is causal, so step $n + 1$ needs only the two 2-vectors $\boldsymbol{\omega}_n, \mathbf{q}_n$. It is also the scheme that survives the extensions of §9: moving shorelines (a nonlinear static solve per step), damping or anelastic memory (extra state variables carried alongside $\boldsymbol{\omega}_n$), and $\mathbf{h}$ or $\boldsymbol{\Gamma}$ (additional terms in $\mathbf{q}$).

8.3 Scheme 2: exact elimination of the rotation (verification and fast path)

Because map (i) of §5 is linear and time-invariant, $\boldsymbol{\omega}$ can be eliminated exactly. This needs no change to `pyslfp`.

1. **Rotational basis (two solves).** For $\mu = 1, 2$, run the *static* generalised solver with no load and rotational forcing $\mathbf{k} = \hat{\mathbf{e}}_\mu$ (§10.1), used here purely as a numerical device. Since $\mathbf{k}$ enters only the rotation row, the resulting fields $X[\hat{\mathbf{e}}_\mu] = (\xi, \mathbf{u}, \phi)$ are those of the rotation $\boldsymbol{\omega}^{(\mu)}$ that the solver returns. With $\mathsf{W} = [\boldsymbol{\omega}^{(1)}\ \boldsymbol{\omega}^{(2)}]$, the static relation $\boldsymbol{\omega} = \mathsf{K}\boldsymbol{\omega} - \mathbf{k}/(I_3 - I_1)$ gives
$$\mathsf{K} = 1 + \frac{\mathsf{W}^{-1}}{I_3 - I_1}, \qquad X^\alpha = \sum_\mu X[\hat{\mathbf{e}}_\mu]\,(\mathsf{W}^{-1})_{\mu\alpha}, \tag{28}$$
where X^α is the full response to a unit prescribed rotation $\boldsymbol{\omega}_\perp = \hat{\mathbf{e}}_\alpha$ with the ocean responding. This is how eq. (21) was computed.

2. **Non-rotating solves.** For each t_n, solve with ζ_n and `rotational_feedbacks=False`. This gives X_n^ζ and $\mathbf{q}_n^\zeta = \beta\boldsymbol{\phi}_{21}^\zeta$. These solves are independent and can be run in parallel.

3. **Rotation.** Integrate eq. (22). With the trapezoidal rule the step is the 2×2 solve $(1 - \mathsf{SK})\boldsymbol{\omega}_{n+1} = \mathsf{R}\boldsymbol{\omega}_n + \mathsf{S}(\mathbf{q}_n + \mathbf{q}_{n+1}^\zeta)$, where $\mathbf{q}_n = \mathbf{q}_n^\zeta + \mathsf{K}\boldsymbol{\omega}_n$. This reproduces Scheme 1 exactly: they agreed to 5×10^{-14} in the test. Alternatively, with $\mathsf{M} = \mathsf{L}(1 - \mathsf{K})$, $\mathbf{f} = -\mathsf{L}\mathbf{q}^\zeta$ taken piecewise linear, $\mathsf{P}_0 = e^{\mathsf{M}\Delta t}$, $\mathsf{P}_1 = \mathsf{M}^{-1}(\mathsf{P}_0 - 1)$ and $\mathsf{P}_2 = \mathsf{M}^{-1}(\mathsf{P}_1/\Delta t - 1)$, the step $\boldsymbol{\omega}_{n+1} = \mathsf{P}_0\boldsymbol{\omega}_n + \mathsf{P}_1\mathbf{f}_n + \mathsf{P}_2(\mathbf{f}_{n+1} - \mathbf{f}_n)$ is exact and has no period error.

4. **Reassemble:** $X_n = X_n^\zeta + \sum_\alpha \omega_{\alpha,n} X^\alpha$.

The cost is two extra solves, and changing the damping, the time step or the initial wobble requires no further sea-level solves. It fails as soon as the problem is nonlinear (moving shorelines) or the response has memory in (i) (viscoelasticity). That is why it is the check rather than the main solver.

8.4 Why not iterate over the whole time history?

The alternative (Method B of the earlier companion note) guesses $\xi(\mathbf{x}, t)$ for all t, builds the excitation history, integrates the wobble ODE over $[0, T]$, recomputes ξ everywhere, and repeats. Its fixed point is the same, but it has two problems.

It diverges in the band of interest. If the propagator uses σ_e, the loop must generate the whole feedback K. One lap then has gain $\kappa/|1 - \nu/\sigma_\mathrm{e}|$ at frequency ν. This exceeds one for periods between $T_e/(1 + \kappa) = 206\,\mathrm{d}$ and $T_e/(1 - \kappa) = 394\,\mathrm{d}$ ($A_\mathrm{w} = A_m$), which contains the annual band. On a finite window the discrete iteration matrix is block lower triangular with spectral radius $\|\mathsf{SK}\| < 1$, so it converges formally. Its non-normal transient growth, of order $(\kappa\sigma_\mathrm{e}T)^k/k!$, is fatal in practice. In a 20-year test with annual forcing the iterates grew by 10^{20} in 35 iterations.

The Chandler-propagator variant is easy to get wrong. Folding the solid-Earth feedback into the propagator, $\dot{\boldsymbol{\omega}} = \sigma_\mathrm{CW}^\mathrm{solid}\mathsf{E}(\boldsymbol{\omega} - \boldsymbol{\chi}')$, is legitimate only with $\boldsymbol{\chi}' = \mathbf{q}_\mathrm{load}^\zeta/(1 - \kappa)$. Here $\mathbf{q}_\mathrm{load}^\zeta$ excludes the tidal term $k_2^T\boldsymbol{\psi}_{21}$, although that term must still be included in the *fields* $\mathbf{u}, \phi$. Using the total $\mathbf{q}$ counts the solid-Earth rotational deformation twice, because $\sigma_\mathrm{CW}\mathsf{E}((1 - \kappa)\boldsymbol{\omega} - \mathbf{q}^\zeta) \neq \sigma_\mathrm{e}\mathsf{E}((1 - \kappa)\boldsymbol{\omega} - \mathbf{q}^\zeta)$. Even when done correctly, the residual ocean gain of about $0.05/|1 - \nu/\sigma_\mathrm{CW}|$ still diverges within about 5% of the Chandler period.

Conclusion. For this linear and causal problem, with its instantaneous response, marching (Scheme 1) is better in every respect. Nonlinearity does not change this, because the problem stays causal and Scheme 1 simply solves a nonlinear static problem at each step.

8.5 Verification checklist

1. **Static limit.** For slowly varying ζ, Scheme 1 tends to the static `pyslfp` solution. Backward Euler with very large Δt reproduces it to solver tolerance.

2. **Free wobble.** With $\gamma = 0$, $\zeta = 0$ and $\boldsymbol{\omega}_0 \neq \mathbf{0}$, the motion is a closed ellipse with period $2\pi/(\sigma_e\sqrt{\det(1-\mathsf{K})})$, as tabulated in §6, with no growth or decay. With $\gamma > 0$ it decays with the chosen Q.

3. K. $\mathsf{K} = \mathsf{K}^T$ and $\operatorname{tr}\mathsf{E}(1-\mathsf{K}) = 0$ to round-off (both confirmed at $l_{\max} = 64$). With $C = 0$, $\mathsf{K} = \kappa 1$.

4. **Scheme 1 = Scheme 2** for the same Δt, to solver tolerance.

5. **Analytic cases.** The ratio of eq. (24) under periodic forcing, and the perpendicular offset of eq. (25) under a trend.

6. **Order of accuracy.** Second-order convergence in Δt.

9 Extensions: where each enters

Relative angular momentum and torques. These enter only through the excitation of eq. (11): replace $\mathbf{q}$ by $\boldsymbol{\chi}$. At leading order this means $\mathbf{q} \to \mathbf{q} + \mathbf{h}_\perp/(I_3 - I_1)$ and $\omega_3 = -(j_3 + h_3)/I_3$. This is AA24's $\mathbf{k}$ with $\mathbf{k} = -\mathbf{h}$, which `pyslfp` already carries as `angular_momentum_change`. A torque enters as $\mathsf{E}\boldsymbol{\Gamma}/[\Omega(I_3 - I_1)]$, rotated by 90°. The terms $\dot{\mathbf{h}}$, and $\dot{\mathbf{j}}$ if wanted, are $O(\epsilon)$ and should be added together or not at all. If they are added, the cleanest state variable is the equatorial angular momentum $\boldsymbol{\ell} = I_1\boldsymbol{\omega} + \mathbf{j} + \mathbf{h}$, for which eq. (10) is exactly

$$\dot{\boldsymbol{\ell}} = \Omega\mathsf{E}\,(I_3\boldsymbol{\omega} - \boldsymbol{\ell}) + \boldsymbol{\Gamma}, \qquad \boldsymbol{\omega} = (\boldsymbol{\ell} - \mathbf{j} - \mathbf{h})/I_1. \tag{29}$$

Scheme 1 carries over directly, marching $\boldsymbol{\ell}$ instead of $\boldsymbol{\omega}$, with $\mathbf{h}$ including $\mathbf{h}_{\text{core}}$ if the core is treated this way rather than through A_w. The inner update $\boldsymbol{\omega} \leftarrow (\boldsymbol{\ell} - \mathbf{j} - \mathbf{h})/I_1$ contracts by about $\delta\kappa \sim 10^{-3}$. Retaining these $O(\epsilon)$ terms is only worthwhile once the larger neglected effects of §4.2 have been modelled.

Anelasticity. The phenomenological damping of eq. (26) is built into the time step. Do not also use an observed or Chandler frequency in the propagator (§8.4). The physical alternative is anelastic, frequency-dependent Love numbers. These give map (i) memory, which in the time domain means internal (relaxation) variables marched alongside $\boldsymbol{\omega}_n$. They would also supply the remaining ≈ 8 d of the Chandler period. Scheme 1 accommodates them, but Scheme 2 does not. Anelastic Love numbers are expected to preserve reciprocity, being a dissipative linear system, but this has not been checked.

Fluid core, beyond leading order. The treatment through A_w (§4) captures the core's main effect. A fuller treatment includes wobble-specific Love numbers and core–mantle coupling (Smith & Dahlen 1981), and ocean and atmospheric transfer functions (Wahr 1982).

Triaxiality. The forward solver needs only eq. (9) with $I_1 \neq I_2$, which gives a 2×2 ODE with different coefficients. The analogues of R and S follow from the same trapezoidal rule. Reciprocity still holds; the adjoint has the rotation reversed and I_1, I_2 exchanged in the gyroscopic term (§10.3).

Observational conventions. See §2. These affect the observation operators, not the forward solver.

10 The generalised problem and reciprocity

This section extends AA24 §§2.7 and 3.2 to the dynamic problem. The aim is that the forward solver of §8, run backwards in time, is also the adjoint solver.

10.1 The generalised dynamic problem

Observables other than sea level need forcings conjugate to each unknown. AA24 add three:

- a surface force density $\mathbf{t}$, paired with $\mathbf{u}$;
- a fictitious surface mass ζ_ϕ, paired with ϕ, which sources the potential but exerts no force;
- a rotational forcing $\mathbf{k}$, paired with $\boldsymbol{\omega}$.

A real load is the combination $(\mathbf{u}' \cdot \nabla\Phi + \phi')\sigma$. All forcings may now depend on time. The generalised dynamic problem is

$$\boxed{\begin{aligned} &A(\mathbf{u}, \phi \,|\, \mathbf{u}', \phi') + \int_{\partial M} (\mathbf{u}' \cdot \nabla\Phi + \phi')\, \sigma \, \mathrm{d}S + \int_{\partial M} (\mathbf{t} \cdot \mathbf{u}' + \zeta_\phi \, \phi') \, \mathrm{d}S \\ &\quad + \int_M \rho \, \mathbf{u}' \cdot \nabla\psi \, \mathrm{d}V + (\tilde{\mathbf{I}}\boldsymbol{\omega} + \mathbf{N}\dot{\boldsymbol{\omega}} - \mathbf{j} + \mathbf{k}) \cdot \boldsymbol{\omega}' = 0 \end{aligned}} \tag{30}$$

with $\boldsymbol{\omega}(0) = \boldsymbol{\omega}_0$. Here σ, ξ and $\int \sigma \, \mathrm{d}S = 0$ are as in eq. (4). Because ζ_ϕ counts as mass for the potential, it also counts in the inertia (AA24 eq. 56):

$$\boldsymbol{\omega}' \cdot \mathbf{j} + \int_M \rho \, \mathbf{u} \cdot \nabla\psi' \, \mathrm{d}V + \int_{\partial M} (\sigma + \zeta_\phi) \, \psi' \, \mathrm{d}S = 0. \tag{31}$$

The matrix N. $\mathbf{N}$ is any matrix with no axial part. This covers three cases. The undamped case with $A_{w,1} = A_{w,2}$ has equatorial block $(A_\mathrm{w}/\Omega)\mathsf{E}$. The triaxial case is eq. (15). The phenomenological damping of eq. (26) has equatorial block $(I_3 - I_1)(\sigma_\mathrm{e}\mathsf{E} + \gamma 1)/(\sigma_\mathrm{e}^2 + \gamma^2)$, which reproduces $\dot{\boldsymbol{\omega}} = (\sigma_\mathrm{e}\mathsf{E} - \gamma 1)(\boldsymbol{\omega} - \mathbf{q})$. In every case the antisymmetric part of $\mathbf{N}$ is gyroscopic and its symmetric part is dissipative.

Correspondence with `pyslfp`. The code supports $\mathbf{t} = \zeta_u \nabla\Phi$ (`displacement_load`), ζ_ϕ (`gravitational_potential_load`) and $\mathbf{k}$ (`angular_momentum_change`). Their responses enter through the generalised Love numbers of AA24 eq. 62. The rotational row is unchanged, with $\mathbf{q} = \beta\boldsymbol{\phi}_{21} - \mathbf{k}_\perp/(I_3 - I_1)$, where $\boldsymbol{\phi}_{21}$ now includes the potential sourced by ζ_ϕ. Scheme 1 therefore solves eq. (30) with the same one-line change as before.

Solvability. For a spherically symmetric reference, the body force $-\rho\nabla\psi$ exerts no net force or torque, so the condition is AA24's eq. 30 at each t: $\int_{\partial M} (\mathbf{t} - \zeta_\phi \nabla\Phi) \cdot (\mathbf{a} + \mathbf{b} \times \mathbf{x}) \, \mathrm{d}S = 0$. Equivalently, the observable must be invariant under rigid-body motions.

10.2 The reciprocity theorem

Let $(\mathbf{u}, \phi, \boldsymbol{\omega})$ solve eq. (30) with loads $(\zeta, \mathbf{t}, \zeta_\phi, \mathbf{k})$ and $\boldsymbol{\omega}(0) = \boldsymbol{\omega}_0$. Let $(\mathbf{u}^\dagger, \phi^\dagger, \boldsymbol{\omega}^\dagger)$ solve it with loads $(\zeta^\dagger, \mathbf{t}^\dagger, \zeta_\phi^\dagger, \mathbf{k}^\dagger)$, with $\mathbf{N}$ replaced by

$$\mathbf{N}^\dagger := -\mathbf{N}^T, \tag{32}$$

and with the *terminal* condition $\boldsymbol{\omega}^\dagger(T) = \mathbf{0}$. Four steps give the theorem.

1. **Instantaneous identity.** Test the first problem with the dagger fields and the second with the undaggered fields, and subtract. The A terms cancel by symmetry, and so do the $\tilde{\mathbf{I}}$ terms. Eliminating $\mathbf{j}$ and $\mathbf{j}^\dagger$ with eq. (31) makes the volume integrals cancel, leaving at each t

$$\int_{\partial M} (\mathbf{u}^\dagger\!\cdot\!\nabla\Phi + \phi^\dagger + \psi^\dagger)\sigma \,\mathrm{d}S + X - \int_{\partial M} (\mathbf{u}\!\cdot\!\nabla\Phi + \phi + \psi)\sigma^\dagger \,\mathrm{d}S - X^\dagger = -\big(\boldsymbol{\omega}^\dagger\!\cdot\!\mathbf{N}\dot{\boldsymbol{\omega}} - \boldsymbol{\omega}\!\cdot\!\mathbf{N}^\dagger\dot{\boldsymbol{\omega}}^\dagger\big), \quad (33)$$

   where $X = \int_{\partial M}[\mathbf{t}\cdot\mathbf{u}^\dagger + \zeta_\phi(\phi^\dagger + \psi^\dagger)]\,\mathrm{d}S + \mathbf{k}\cdot\boldsymbol{\omega}^\dagger$, and $X^\dagger$ is the same with the roles exchanged.

2. **Total derivative.** By eq. (32), $\boldsymbol{\omega}\cdot\mathbf{N}^\dagger\dot{\boldsymbol{\omega}}^\dagger = -\dot{\boldsymbol{\omega}}^\dagger\cdot\mathbf{N}\boldsymbol{\omega}$. The right-hand side of eq. (33) is therefore $-\frac{\mathrm{d}}{\mathrm{d}t}(\boldsymbol{\omega}^\dagger\cdot\mathbf{N}\boldsymbol{\omega})$, for any $\mathbf{N}$. This is why the adjoint carries $-\mathbf{N}^T$. When $\mathbf{N}$ is antisymmetric, $\mathbf{N}^\dagger = \mathbf{N}$.

3. **Integrate over $[0, T]$.** The boundary term is $-[\boldsymbol{\omega}^\dagger\cdot\mathbf{N}\boldsymbol{\omega}]_0^T = \boldsymbol{\omega}^\dagger(0)\cdot\mathbf{N}\boldsymbol{\omega}_0$, since $\boldsymbol{\omega}^\dagger(T) = \mathbf{0}$.

4. **Convert to sea level.** Use eq. (4). The constants $\Phi_g, \Phi_g^\dagger$ vanish against $\int\sigma^\dagger = \int\sigma = 0$, and the ocean terms $\rho_w\int C\xi\xi^\dagger$ cancel. Finally divide by $-g$.

The result is

$$\begin{aligned}\int_0^T \Big\{ \int_{\partial M} \xi^\dagger\zeta \,\mathrm{d}S - \frac{1}{g}\int_{\partial M} \big[\mathbf{t}\cdot\mathbf{u}^\dagger + \zeta_\phi(\phi^\dagger + \psi^\dagger)\big]\,\mathrm{d}S - \frac{1}{g}\,\mathbf{k}\cdot\boldsymbol{\omega}^\dagger \Big\}\,\mathrm{d}t \\ = \int_0^T \Big\{ \int_{\partial M} \xi\,\zeta^\dagger \,\mathrm{d}S - \frac{1}{g}\int_{\partial M} \big[\mathbf{t}^\dagger\cdot\mathbf{u} + \zeta_\phi^\dagger(\phi + \psi)\big]\,\mathrm{d}S - \frac{1}{g}\,\mathbf{k}^\dagger\cdot\boldsymbol{\omega} \Big\}\,\mathrm{d}t - \frac{1}{g}\,\boldsymbol{\omega}^\dagger(0)\cdot\mathbf{N}\boldsymbol{\omega}_0. \quad (34)\end{aligned}$$

This is AA24's eq. 57, integrated over time, plus an initial-state term. With only direct loads and $\boldsymbol{\omega}_0 = \mathbf{0}$ it reduces to

$$\boxed{\int_0^T\!\!\int_{\partial M} \xi^\dagger\,\zeta \,\mathrm{d}S\,\mathrm{d}t = \int_0^T\!\!\int_{\partial M} \xi\,\zeta^\dagger \,\mathrm{d}S\,\mathrm{d}t} \quad (35)$$

Since $\psi = -[\mathbf{x}\times(\boldsymbol{\Omega}\times\mathbf{x})]\cdot\boldsymbol{\omega}$, the $\zeta_\phi\psi$ terms can be moved into the rotational forcing by replacing $\mathbf{k}$ with $\mathbf{k} - \int_{\partial M}\zeta_\phi\,\mathbf{x}\times(\boldsymbol{\Omega}\times\mathbf{x})\,\mathrm{d}S$ (AA24 eq. 58). This leaves ζ_ϕ paired with ϕ alone.

10.3 The adjoint problem in practice

Write $\tau = T - t$ and $\tilde{\boldsymbol{\omega}}(\tau) = \boldsymbol{\omega}^\dagger(T - \tau)$, and similarly for the other fields and loads. Then $\mathbf{N}^\dagger\dot{\boldsymbol{\omega}}^\dagger = -\mathbf{N}^T\dot{\boldsymbol{\omega}}^\dagger = \mathbf{N}^T\,\mathrm{d}\tilde{\boldsymbol{\omega}}/\,\mathrm{d}\tau$, and the adjoint problem becomes

$$\text{eq. (30) in } \tau, \text{ with } \mathbf{N}\to\mathbf{N}^T, \text{ loads } (\tilde{\zeta}, \tilde{\mathbf{t}}, \tilde{\zeta}_\phi, \tilde{\mathbf{k}}), \ \tilde{\boldsymbol{\omega}}(0) = \mathbf{0}. \quad (36)$$

In words: *run the forward generalised solver forwards in τ, on time-reversed adjoint loads, with $\mathbf{N}$ replaced by its transpose, then reverse the output.*

Case	$\mathbf{N}^T$ in terms of $\mathbf{N}$	Adjoint solver
$A_{w,1} = A_{w,2}$, undamped	$-\mathbf{N}$	rotation reversed ($\sigma_\mathrm{e}\to-\sigma_\mathrm{e}$)
damped	gyroscopic part reversed, damping kept	rotation reversed, same γ
triaxial	$-\mathbf{N}$ with $A_{w,1}\leftrightarrow A_{w,2}$	reversed, $A_{w,1}, A_{w,2}$ swapped

With damping, the adjoint is therefore as stable in τ as the forward problem is in t. The static parts ($\tilde{\mathbf{I}}$, K, the Love numbers and the ocean) are self-adjoint and unchanged.

In Scheme 1. The step matrices R and S are rational functions of the matrix $\mathsf{L} = \sigma_\mathrm{e}\mathsf{E} - \gamma\mathsf{1}$. Replacing L by L^T replaces them by R^T and S^T, which for $\gamma = 0$ is $a\to -a$. The adjoint solver is Scheme 1 with $\mathsf{R}^T, \mathsf{S}^T$, run on reversed loads.

Green's function. Let $G_{\mathbf{N}}(\mathbf{x}_2, \mathbf{x}_1; s)$ be the sea level at $\mathbf{x}_2$ a time s after a unit impulsive load at $\mathbf{x}_1$. Then eq. (35) gives

$$G_{\mathbf{N}}(\mathbf{x}_2, \mathbf{x}_1; s) = G_{\mathbf{N}^T}(\mathbf{x}_1, \mathbf{x}_2; s). \tag{37}$$

Swapping source and receiver is compensated by reversing the rotation, which is Onsager–Casimir reciprocity. Integrating over s recovers AA24's static symmetry, because the static response does not depend on $\mathbf{N}$.

10.4 Sensitivity kernels and adjoint loads

Suppose a functional J has first-order perturbation

$$\delta J = \int_0^T \Big\{ \int_{\partial M} \zeta^\dagger \delta\xi \,\mathrm{d}S - \frac{1}{g} \int_{\partial M} \left[\mathbf{t}^\dagger \cdot \delta\mathbf{u} + \zeta_\phi^\dagger (\delta\phi + \delta\psi) \right] \mathrm{d}S - \frac{1}{g}\, \mathbf{k}^\dagger \cdot \delta\boldsymbol{\omega} \Big\}\, \mathrm{d}t. \tag{38}$$

Apply eq. (34) with a forward problem driven by $\delta\zeta$ and $\delta\boldsymbol{\omega}_0$ only. This gives

$$\delta J = \int_0^T \int_{\partial M} \xi^\dagger \, \delta\zeta \,\mathrm{d}S \,\mathrm{d}t + \frac{1}{g}\, \boldsymbol{\omega}^\dagger(0) \cdot \mathbf{N}\, \delta\boldsymbol{\omega}_0. \tag{39}$$

The ice kernel is $K(\mathbf{x}, t) = \rho_i (1 - C)\, \xi^\dagger(\mathbf{x}, t)$. The second term is the kernel for the initial wobble, which an inversion must treat as a nuisance parameter unless the run starts from rest (§5). The adjoint loads for common observables follow AA24 §4 and the altimetry row of Note 1, each multiplied by a time weight $w(t)$ ($\delta(t - t_o)$ at an instant, a boxcar for a monthly mean):

Observable (times $w(t)$)	$\zeta^\dagger$	$\mathbf{t}^\dagger$	$\zeta_\phi^\dagger$	$\mathbf{k}^\dagger$
$\xi(\mathbf{x}_0)$	$\delta(\mathbf{x}, \mathbf{x}_0)$	$\mathbf{0}$	0	$\mathbf{0}$
$\hat{\mathbf{n}} \cdot \mathbf{u}(\mathbf{x}_0)$	0	$-g\, \delta(\mathbf{x}, \mathbf{x}_0) \hat{\mathbf{n}}$	0	$\mathbf{0}$
ϕ_{lm} (AA24 eq. 66)	0	$\mathbf{0}$	$-g\, Y_{lm}$	$\int_{\partial M} \zeta_\phi^\dagger\, \mathbf{x} \times (\boldsymbol{\Omega} \times \mathbf{x})\, \mathrm{d}S$
$\int_{\partial M} v\, \Delta\mathrm{SSH}^*\, \mathrm{d}S$	v	$-g\, v\, \hat{\mathbf{n}}$	0	$\int_{\partial M} v\, \mathbf{x} \times (\boldsymbol{\Omega} \times \mathbf{x})\, \mathrm{d}S$
$\hat{\mathbf{e}}_\alpha \cdot \boldsymbol{\omega}$ (polar motion)	0	$\mathbf{0}$	0	$-g\, \hat{\mathbf{e}}_\alpha$

In the fourth row, v is the spatial averaging weight and $\Delta\mathrm{SSH}^* = \xi + \hat{\mathbf{n}} \cdot \mathbf{u} + \psi/g$. This is the geocentric sea-surface height with only the direct centrifugal term removed (Lickley et al. 2018). It is a modelling convention, not the operational pole-tide correction. For products corrected in the standard way, eq. (2) applies instead. The operator is then the non-rotating one, whose adjoint has $\zeta^\dagger = v$ and $\mathbf{t}^\dagger = -gv\hat{\mathbf{n}}$, solved with rotational feedback switched off (`rotational_feedbacks=False`). Setting $\mathbf{k}^\dagger = \mathbf{0}$ in the rotating adjoint is not enough, since that retains the load-driven rotation that cancels from the data. The $\mathbf{k}^\dagger$ entries in rows three and four are non-zero only for $(l, m) = (2, \pm 1)$, plus $l = 0$ and $(2, 0)$ if ω_3 is kept.

Causality. If $w(t) = 0$ for $t > t_o$, the adjoint fields vanish for $t > t_o$: in τ nothing forces them until the observation time is reached. The kernel therefore automatically ignores loads applied after the measurement. Before t_o the kernel contains a free wobble at the Chandler period, decaying into the past with an e-folding time of about 40 yr for $Q \approx 100$, and not at all if $\gamma = 0$ (§10.6).

10.5 Discrete reciprocity

The trapezoidal scheme satisfies reciprocity *exactly*, with no separately derived discrete adjoint, provided time integrals are evaluated by pairing interval means:

$$\int_0^T f\, g \,\mathrm{d}t \;\to\; \sum_n \Delta t\, \bar{f}_{n+\frac{1}{2}}\, \bar{g}_{n+\frac{1}{2}}, \qquad \bar{f}_{n+\frac{1}{2}} = \tfrac{1}{2}(f_n + f_{n+1}). \tag{40}$$

The reason is that the trapezoidal step is exactly eq. (30) applied to interval means, with $\dot{\boldsymbol{\omega}}$ replaced by $(\boldsymbol{\omega}_{n+1} - \boldsymbol{\omega}_n)/\Delta t$. The instantaneous relations are linear, so the means satisfy them exactly. The gyroscopic terms telescope for any $\mathbf{N}$:

$$\bar{\boldsymbol{\omega}}^\dagger \cdot \mathbf{N}(\boldsymbol{\omega}_{n+1} - \boldsymbol{\omega}_n) - \bar{\boldsymbol{\omega}} \cdot \mathbf{N}^\dagger(\boldsymbol{\omega}_{n+1}^\dagger - \boldsymbol{\omega}_n^\dagger) = \boldsymbol{\omega}_{n+1}^\dagger \cdot \mathbf{N}\boldsymbol{\omega}_{n+1} - \boldsymbol{\omega}_n^\dagger \cdot \mathbf{N}\boldsymbol{\omega}_n.$$

This was confirmed on the reduced coupled model (the K of eq. (21), a random symmetric non-rotating response, and the rotational coupling implied by §6.2). The two sides of eq. (35) agreed to 10^{-14}, with and without damping, at every Δt. Pairing nodal values with trapezoidal weights instead agrees only to $O(\Delta t^2)$. In practice, define observables as interval means, which is natural for monthly GRACE and altimetry products. Adjoints from the same solver then pass dot-product tests at machine precision.

10.6 Things to be aware of

1. **Untested in the full code.** Every numerical test used the reduced model built from a real K. Scheme 1 has not yet been implemented in `pyslfp`. Its combined ocean-plus-rotation inner iteration is argued to converge but has not been demonstrated.

2. **Near-resonant results depend on the Chandler period.** With $A_\mathrm{w} = A_m$ the model gives 422 d, against 425.5 d for a complete elastic calculation and about 433 d observed. The annual prograde factor of eq. (24) is nevertheless -6.4 at 422 d and -5.4 at 433 d. Annual-band rotational corrections therefore carry an uncertainty of order 20% unless the remaining anelastic lengthening is modelled. Trends are unaffected.

3. **Long memory.** The elastic free wobble never decays. Any excitation (a load switched on, an inconsistent initial state) then persists for ever, and adjoint kernels extend over the whole past at full amplitude. This is why $Q \approx 100$ damping is the recommended default (§8.1). Even then the e-folding time is about $QT_\mathrm{CW}/\pi \approx 40\,\mathrm{yr}$, so $\boldsymbol{\omega}_0$ and loads before the window genuinely matter.

4. **Real polar motion is not load-driven.** The observed wobble is excited mainly by the atmosphere and ocean, and its pole tide (a few mm) far exceeds the load-induced one. For corrected data this is removed by processing (§2). For uncorrected data it is signal the model does not contain.

A Audit of the earlier notes

Checked and correct. Restatement of AA24 §§2–3 (including the Euler-equation algebra, eqs. 37–49, and the MacCullagh factors). The Liouville equation D3.2 and its equatorial form D3.3, re-derived and checked symbolically here. The gyroscopic matrix and its antisymmetry condition. The size estimates in the ordering argument. The axial result. DP1–DP2, DP5 (444 d) and $k_s = 0.951$, $\kappa = 0.314$ (recomputed from the `pyslfp` constants). The periodic-response table. The magnitude of the trend offset (70.7 d). The trapezoidal and backward-Euler matrices, $\mathsf{R} + 2\mathsf{S} = 1$, the Scheme A contraction bound, and the 450.6 d discrete period (450.8 d here). The time-integrated reciprocity, terminal condition and time reversal. The consistency requirements 1 and 2 of the companion note's Method B.

Issues.

1. **Method B double counts the solid-Earth feedback (companion note, Steps 3–4 and requirement 3).** The "recommended" variant pairs a no-ocean Chandler propagator with an excitation read from the *total* potential, which already contains $k_2^T \boldsymbol{\psi}_{21}$. It also omits the factor $1/(1-\kappa)$. The correct pairing is given in §8.4. Since whole-history iteration is not recommended anyway, this affects no conclusion.

2. **The $\dot{\mathbf{j}}$ refinement (Note 1 §3.3; companion §2.7)** is algebraically right and harmless, but not systematically better. It is an $O(\epsilon)$ correction to the excitation (3% monthly, 0.3% annual), of the same order as the neglected relative angular momentum of the transported meltwater, and far smaller than the fluid-core and anelastic effects. Standard excitation functions omit it. The reciprocity discussion built on it (D3.11–D3.12) is then unnecessary. The step-load jump it predicts (-0.23% of the static response) is real but negligible.

3. **The trend offset is not a lag** (Note 1 DP7 and its gloss). The offset is perpendicular to the trend, a sideways displacement of the pole path, not a delay along it (§7).

4. **The fluid core** is treated in both notes as a later refinement, with the suggestion of tuning σ_e. Its relative angular momentum is small, but its gyroscopic term is about 11% of the retained inertial term, the largest effect outside the elastic theory. It is included here through $A_w = A_m$ (§4), which brings the model period within 1% of a complete elastic calculation.

5. **Pole-tide conventions.** Both notes treat observational conventions as a side issue, and Note 1's altimetry row uses the direct-term convention $\Delta\mathrm{SSH}^*$. For data corrected in the standard way, the load-driven rotation cancels from the observables (§2).

6. **Damping and adjoints (Note 1, "Physics not included").** The statement that the adjoint "carries damping of the opposite sign" is misleading. In reversed time the adjoint is equally damped, and rotating-system reciprocity survives (§10).

7. **Triaxiality and discrete adjoints (Note 1).** Note 1 said reciprocity was not established for $I_1 \neq I_2$, and recommended transposing the time-step recursion to get an exact discrete adjoint. Both are superseded. With $\mathbf{N}^\dagger = -\mathbf{N}^T$, reciprocity holds for any $\mathbf{N}$ (§10.2). With interval-mean pairing, the trapezoidal solver is its own exact discrete adjoint (§10.5).

8. **Symmetry of K** was conjectured in Note 1 but not proved. It is proved in §6.2 and confirmed numerically. The ocean lengthens T_{CW} from 444 to 476 d.

9. **Divergence of the global iteration** (companion §3.3). The frequency-domain argument is a heuristic. On a finite window the iteration converges formally but with catastrophic transient growth (§8.4). The conclusion is unchanged.

10. **Scheme B** in both notes asked for a prescribed-$\boldsymbol{\omega}$ solve, which `pyslfp` lacks. Eq. (28) obtains the same basis from the existing angular-momentum forcing.

11. **Phase drift.** The notes quote the per-cycle period error of the trapezoidal rule but not its accumulation over long runs (§8.1).

12. **Sign conventions.** The companion note's eq. B4 uses the opposite sign for ϕ and ψ. Everything here follows AA24.

13. **Constants.** `pyslfp` uses $\Omega = 2\pi/86400\,\mathrm{s}^{-1}$ (solar day), not the sidereal $7.2921 \times 10^{-5}\,\mathrm{s}^{-1}$. This changes σ_e by 0.27% and k_s, κ and K by 0.55%, which is small but should be corrected in the code.

14. **Notation.** Note 1 used $\mathbf{i}$ for the inertia perturbation, $\boldsymbol{\Gamma}$ for the gyroscopic matrix, $\Delta\mathrm{SL}$ and $\mathcal{C}$. These are now $\mathbf{c}$, $\mathbf{N}$, ξ and C, and $\boldsymbol{\Gamma}$ is the external torque.

Omitted as unnecessary for now. The demonstration design of the companion note (§4) and its $\dot{\mathbf{j}}$-based refinements. Neither is wrong as a plan, and the demonstration can be revisited.

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